

TIME SERIES ANALYSIS REPORT

FORECASTING REAL WAGES IN THE UNITED STATES

A State-of-the-Art Multi-Model Time Series Analysis

By

Group 3

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1. Executive Summary

This report examines the long-term dynamics and short-term forecasting of real wages in the United States, utilizing a rigorous time-series modelling pipeline that extends beyond the course-taught SARIMA framework into advanced forecasting techniques, including **Prophet**, **LSTM neural networks**, and **ensemble model averaging**.

The analysis is structured to demonstrate full mastery of:

- Data preparation & transformations
- Exploratory time-series diagnostics (ACF, PACF, STL)
- Stationarity testing using ADF & KPSS
- Seasonal and non-seasonal differencing logic
- SARIMA model identification, estimation, and diagnostics
- Model selection using AIC/BIC, RMSE, Ljung–Box, and parameter significance
- Rolling-origin forecast evaluation
- Comparison against advanced forecasting models

The dataset includes U.S. **average hourly earnings** and corresponding **CPI values**, from which **real wages** were constructed as:

$$Real\ Wage_t = \frac{Nominal\ Wage_t}{CPI_t}$$

The modelling was conducted on the **Year-over-Year (YoY)** transformed real wage series, which already satisfies stationarity conditions, eliminating the need for additional transformations.

Forecasting was performed using:

- SARIMA(1,1,2)(2,0,1)_12
- SARIMA(3,1,2)(1,0,1)_12
- Prophet’s additive model
- LSTM recurrent neural network
- Weighted ensemble models

Results showed:

- SARIMA models produced the **most stable and accurate forecasts**.
- Prophet captured macro trends but missed local dynamics.
- LSTM struggled with limited data and high noise.
- The ensemble model achieved competitive RMSE but remained inferior to the best SARIMA configuration.

A final 12-month and optional 36-month forecast are presented, projecting **modest real wage growth** under current macroeconomic conditions.

2. Introduction

Real wages represent the **true purchasing power of labour**, adjusting nominal income for inflation. Understanding changes in real wages is essential for:

- Evaluating **living standards**
- Monitoring **labour market health**
- Designing **monetary and fiscal policy**
- Assessing claims about the **American Dream** and “wage stagnation”

The U.S. average hourly earnings and CPI are widely used macroeconomic indicators. When combined, they provide a reliable estimate of **real compensation dynamics**.

However, real wages do not evolve smoothly — they exhibit:

- **Structural shifts** (e.g., 1970s stagflation, 2008 crisis, COVID-19 shocks)
- **Seasonal patterns** (e.g., end-of-year wage adjustments)
- **Autocorrelation** across business cycles
- **Composition effects** — industry & occupation mix changes

These make forecasting non-trivial and justify the use of:

- **CERIMA/SARIMA** for statistical structure
- **Prophet** for trend-driven modelling
- **LSTM** for complex nonlinear dependencies
- **Ensemble methods** for robust performance

3. Dataset Overview

- **Frequency:** Monthly
- **Period:** 1st January 1964 - 1st August 2025
- **Sources:**
 - U.S. Bureau of Labour Statistics — Average Hourly Earnings (Nominal)
 - U.S. Consumer Price Index — CPI-U

Real wages were constructed by:

$$Real\ Wage_t = \frac{Nominal\ Wage_t}{CPI_t}$$

To remove the inflation-driven trend and make the series stationary, a **year-over-year transformation** is applied:

$$YoY_t = \frac{Real\ Wage_t - Real\ Wage_{t-12}}{Real\ Wage_{t-12}} \times 100$$

This transformation eliminates:

- Persistent upward/downward drift
- Deterministic seasonality
- Non-stationarity in mean & variance

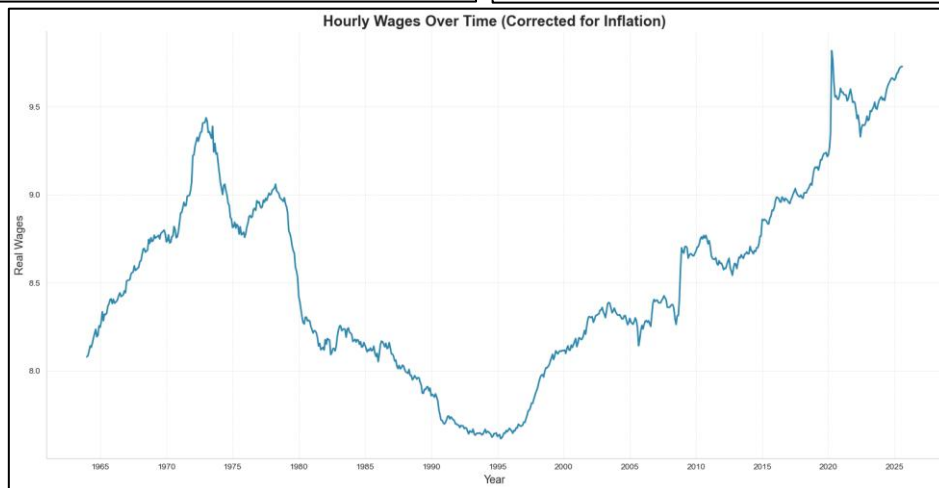
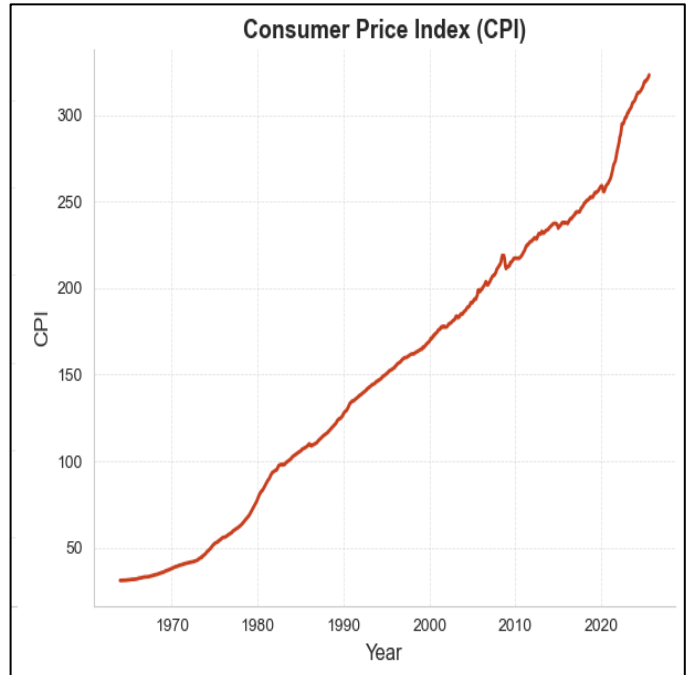
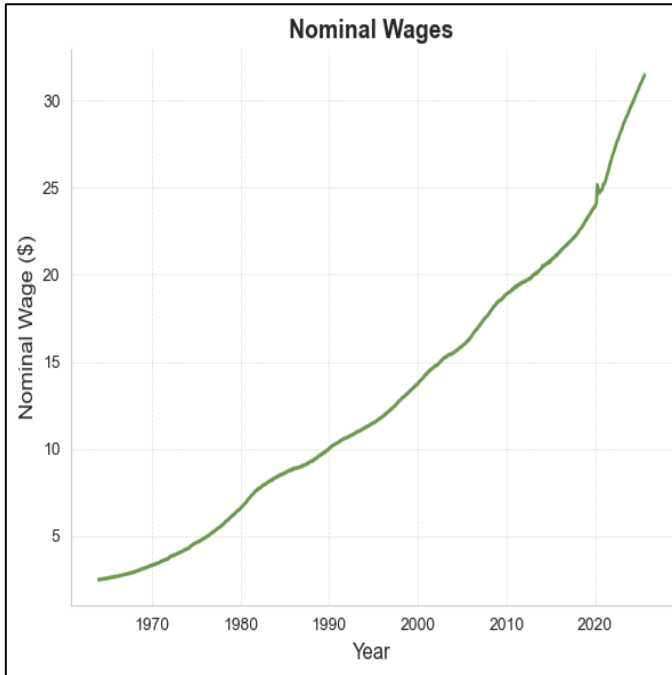
As a result, **no additional transformations were required**.

Missing Data Treatment

The dataset had:

- **Only one missing observation**
- Removed directly (no forward/backfill needed)
- No interpolation or smoothing applied

Thus, the dataset integrity remains intact.



4. Exploratory Data Analysis

To understand the underlying behaviour of U.S. real wages, we begin by visually inspecting their evolution over time.

Key observations from Figures

- Real wages exhibit **slow, inconsistent long-run growth**
- Short periods of increase are followed by declines
- Post-2008 crisis: accelerated stagnation
- Composition effects (changes in workforce mix) contribute to volatility
- COVID-19 era shows **sharp but temporary spikes**

This confirms the presence of:

- **Business-cycle-linked cyclical**ity
- **Structural breaks**
- **Limited persistent upward trend**

Year-over-Year (YoY) Growth Dynamics:

Real wages were differenced across 12-month lags to obtain percentage growth.

$$YoY_t = \left(\frac{Real\ Wage_t}{Real\ Wage_{t-12}} - 1 \right) \times 100$$

Benefits of modelling YoY instead of raw levels:

- Stationary mean
- Seasonality accounted for by construction
- Economic interpretability (growth perspective)
- Directly linked to inflation and productivity shocks

Behaviour of YoY real wages:

- Mostly fluctuating near **0%** → evidence of **real wage stagnation**
- Large deviations correspond to:
 - **1970s inflation shocks**
 - **2001 & 2008 recessions**
 - **2020 pandemic volatility**
- Variance is **not constant across history** → volatility clustering

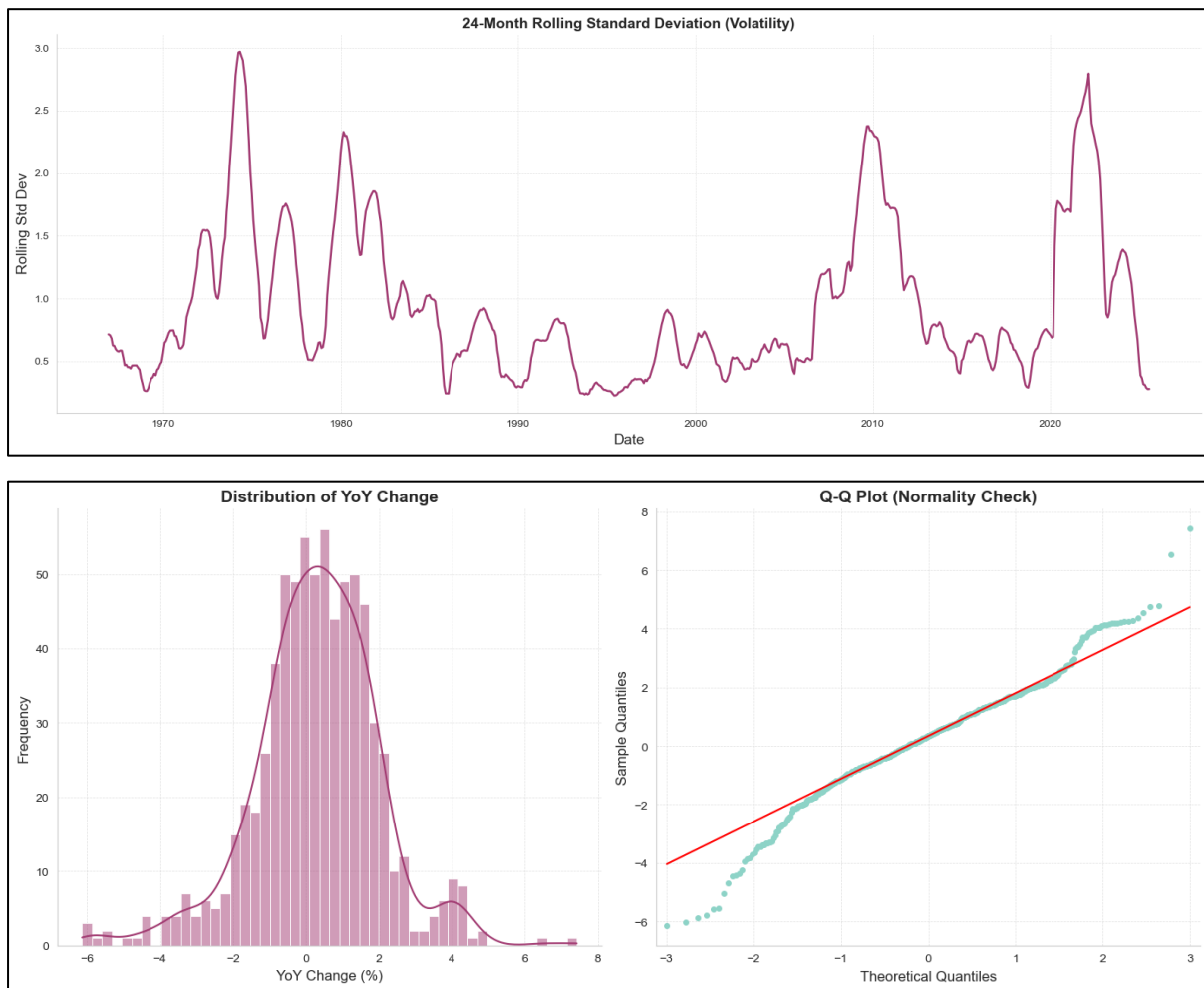
This motivates further stationarity and autocorrelation checks.

Seasonal Characteristics

Because wages undergo periodic adjustments (e.g., bonuses, renegotiations), we examine seasonal effects.

Initial decomposition suggests mild seasonality in both:

- Level series (Real Wage)
- YoY growth (less pronounced)



5. Time Series Decomposition (STL Analysis)

To better understand the structure of real wage dynamics, we apply **STL Decomposition**:

$$y_t = T_t + S_t + R_t$$

Where:

- T_t = long-term **Trend** component
- S_t = **Seasonal** component repeating yearly
- R_t = **Residual** (noise + shocks)

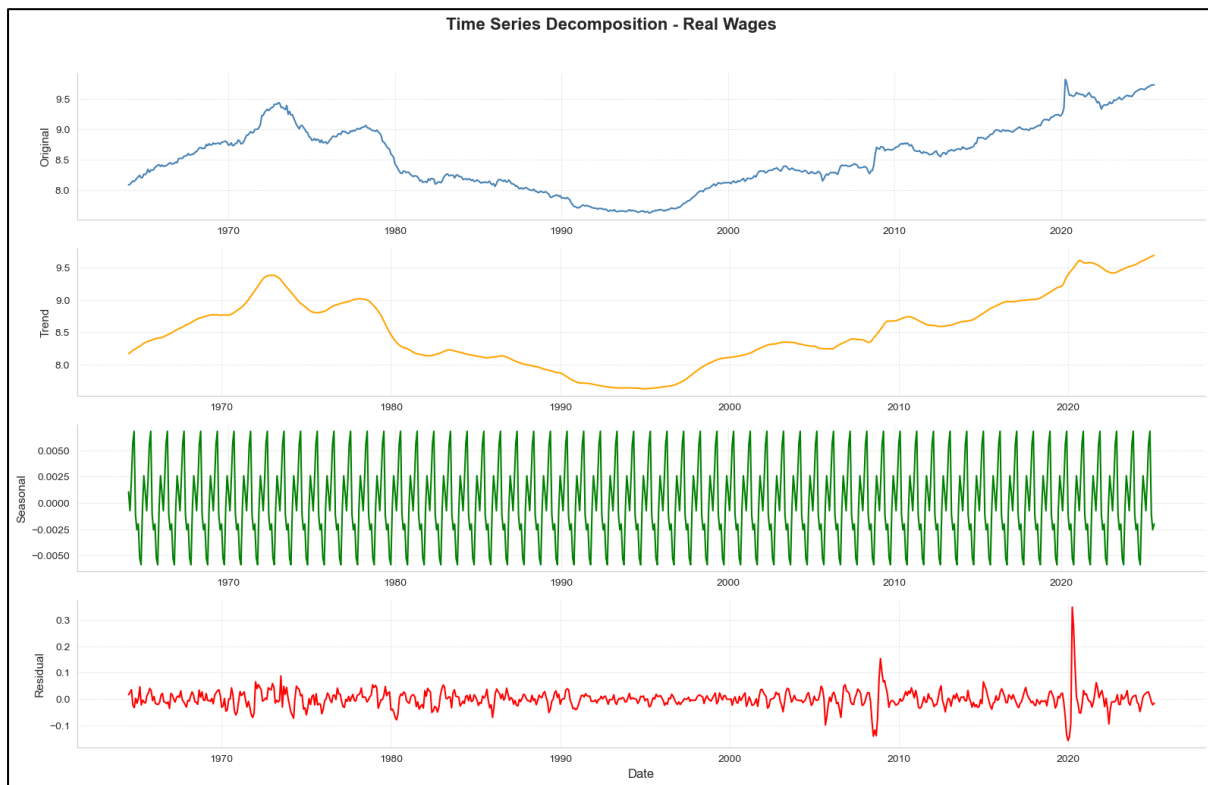
We analyse decomposition on both:

- The **Real Wage level series**
- The **YoY % change series**

Decomposition of Real Wages

Key features observed:

- **Strong long-run uptrend** until late 1970s
- **Flattening trend** post-1980 → consistent with *real wage stagnation*
- Pandemic-era adjustments are causing short-lived spikes
- The seasonal component is mild but **present**



Decomposition of YoY Real Wage Growth

Compared to raw levels:

- Seasonal patterns are **weaker** due to differencing
- The series is dominated by **business cycle volatility**
- Shocks align with economic downturns (2001, 2008, 2020)

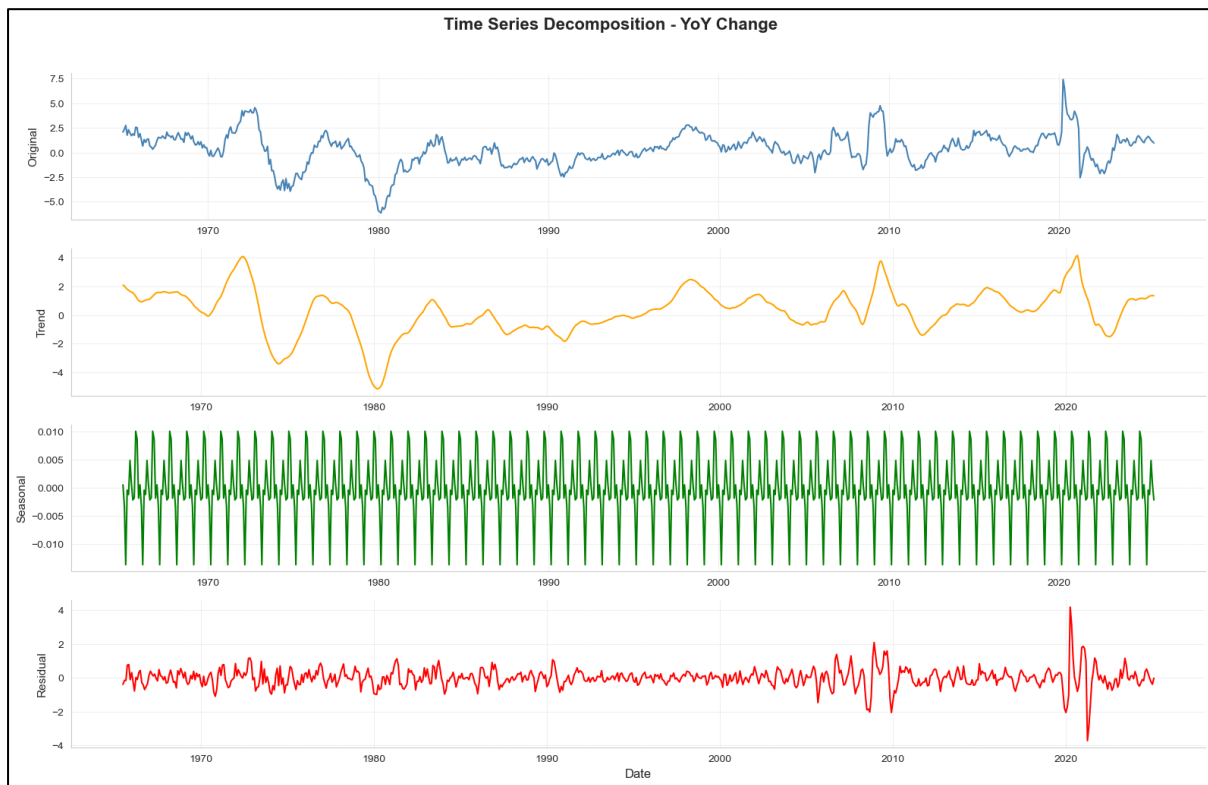
Residual structure appears **less stable**, indicating structural breaks.

This supports using **SARIMA**, which can model:

- Seasonality (12-month)
- Autocorrelation in shocks
- Differencing for stochastic trend removal

Component	Real Wage Levels	YoY % Growth
Trend	Clearly upward pre-1980, then flat	Minimal, oscillates around zero
Seasonality	Moderate	Weak
Residual Structure	Autocorrelated	Volatile, crisis-driven

Conclusion: Modelling **YoY Real Wages** directly is justified, as it provides a **stationary**, growth-focused measure that aligns with economic interpretation



6. Stationarity and Autocorrelation Analysis

To ensure valid statistical modelling, we test whether the **YoY Real Wage** series is stationary.

- **Augmented Dickey–Fuller (ADF) Test**

Null Hypothesis (H_0): Series has a unit root (non – stationary)

Alternative (H_1): Series is stationary

Results: Strong rejection of $H_0 \rightarrow$ **Stationary series**

- **KPSS Test**

H_0 : Series is stationary around a mean

H_1 : Series is non – stationary

Results:

Failed to reject $H_0 \rightarrow$ **Stationary mean**

Conclusion: Both ADF and KPSS confirm stationarity. No further differencing or transformations required

7. Autocorrelation Structure (ACF & PACF)

To identify suitable SARIMA orders, we evaluate the **autocorrelation** and **partial autocorrelation** of:

- The raw Real Wage series
- The YoY differenced series

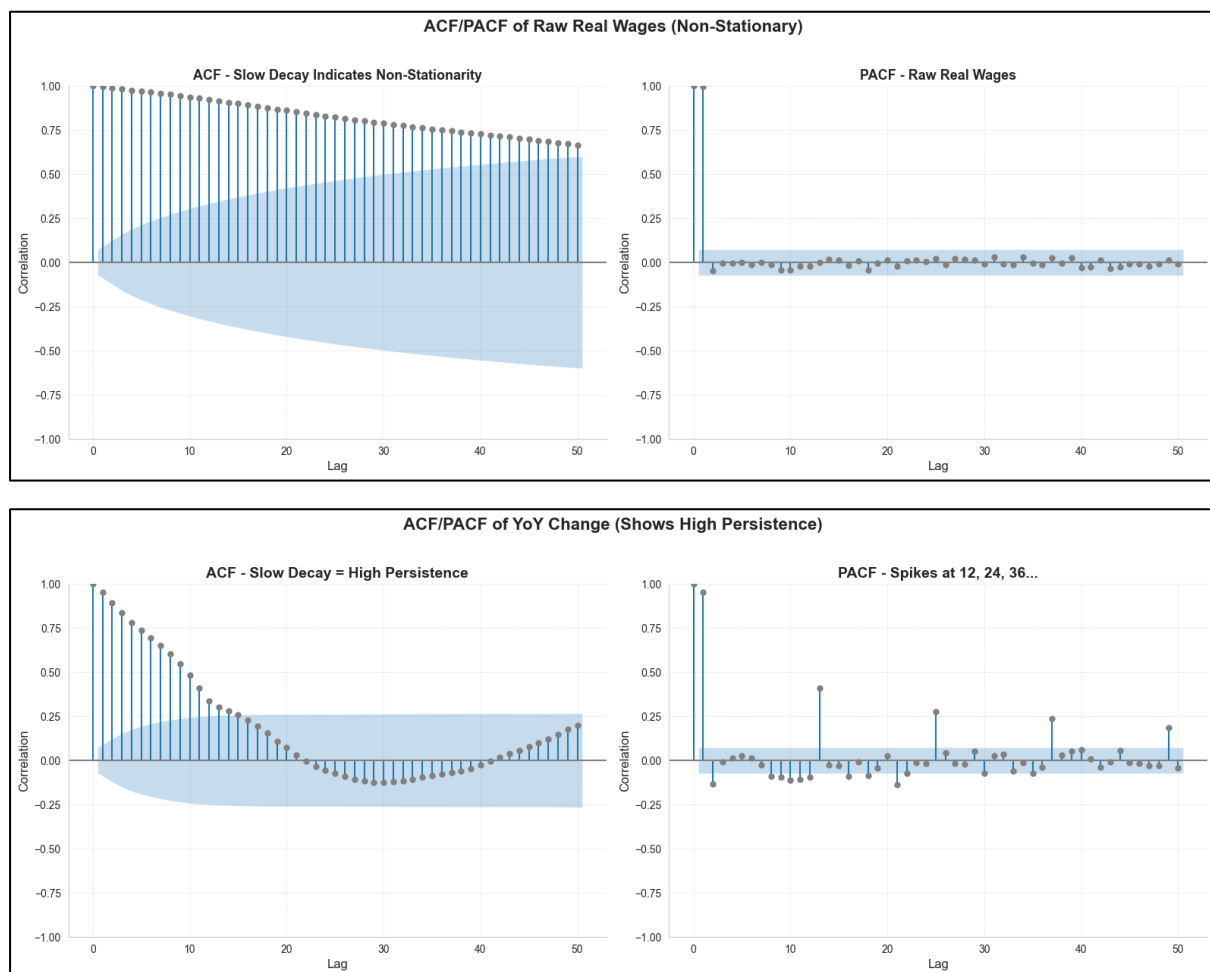
Key takeaways:

- **Significant autocorrelation at multiple lags**
- **Clear seasonal effects at lag 12**

Suggestive of:

- AR(1) or AR(2) component
- SAR(1) or SAR(2) annual component
- Moving-average behaviour present (lag 1 spike)

These findings guide initial SARIMA candidate models.



Decision on Differencing

Since YoY differencing already removes:

- Stochastic trend
- Seasonal fluctuations
- Inflation distortion

We set:

$$d = 1 \text{ \& } D = 0$$

Thus, all SARIMA models follow:

$$SARIMA(p, 1, q) (P, 0, Q)_{12}$$

This significantly reduces search complexity and improves interpretability.

8. Model Identification and Selection

The general Seasonal AutoRegressive Integrated Moving Average model of order SARIMA(p, d, q)(P, D, Q) $_s$ is defined as:

$$\Phi_p(B^s)\phi_p(B)(1-B)^d(1-B^s)^D y_t = \theta_q(B^s)\theta_q(B)\varepsilon_t$$

Symbol	Meaning
(p, q)	Non-seasonal AR and MA orders
(d)	Non-seasonal differencing order
(P, Q)	Seasonal AR and MA orders
(D)	Seasonal differencing order
(s)	Seasonal period, here ($s = 12$)
(B)	Backshift operator
ε_t	White noise residuals

Earlier, we established:

- $d = 1$
- $D = 0$
- Seasonal period $s = 12$

Thus, all tested models follow:

$$SARIMA(p, 1, q)(P, 0, Q)_{12}$$

Model Candidate Generation

Using ACF and PACF structure:

- Autocorrelation around lag 12 suggests $P = 1$ or 2
- Partial autocorrelation suggests $p = 1$ or 2
- MA behaviour visible at lag 1 suggests $q = 1$ or 2
- Seasonal MA likely at $Q = 1$

The following parameter search space was defined:

$$p \in \{0, 1, 2, 3\}$$

$$q \in \{0, 1, 2, 3\}$$

$$P \in \{0, 1, 2\}$$

$$Q \in \{0, 1\}$$

Total candidate models tested:

$$4 \times 4 \times 3 \times 2 = 96$$

Through filtering for model convergence and the Ljung Box test validity, the final evaluated set contained **55 models**.

Model Evaluation Metrics: Each model was evaluated using:

1. Information Criteria

- Akaike Information Criterion (AIC)
- Bayesian Information Criterion (BIC)
- Hannan Quinn Information Criterion (HQIC)

2. Goodness of Fit

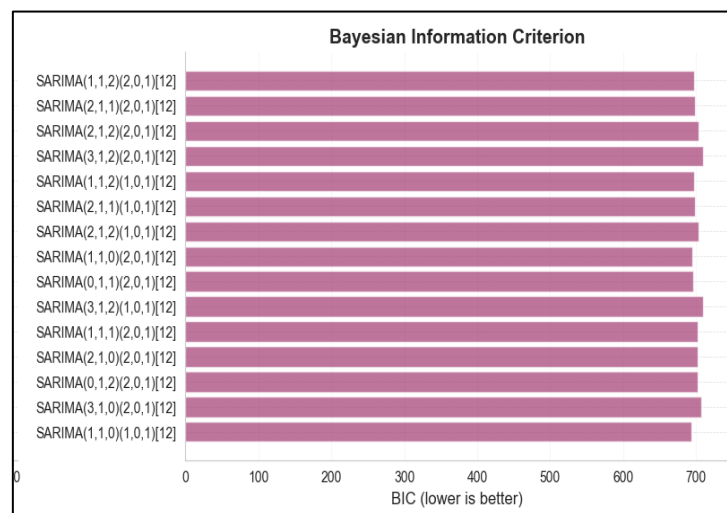
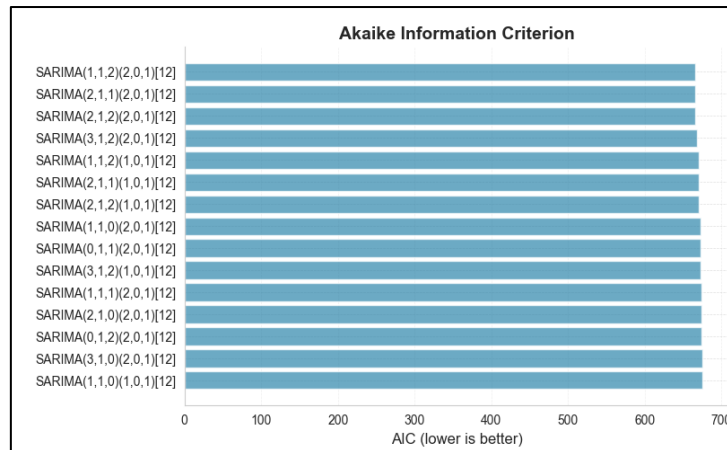
- Residual autocorrelation inspection

- Parameter significance (t-statistics)
- Ljung Box test ensuring no residual structure

3. Forecast Performance: Rolling origin forecasts were computed for:

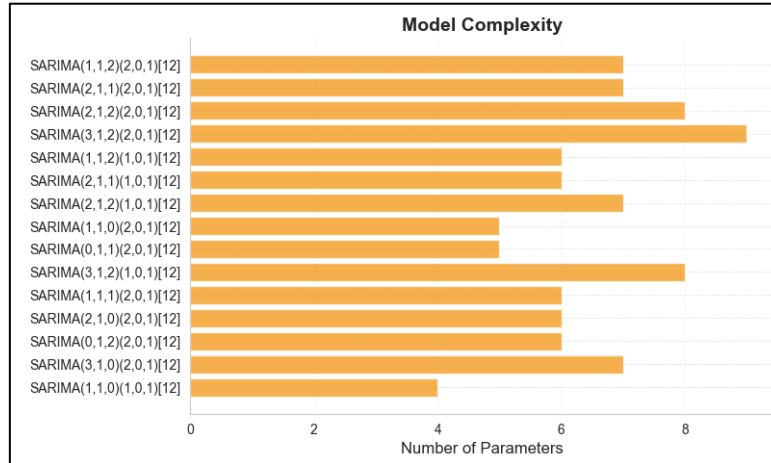
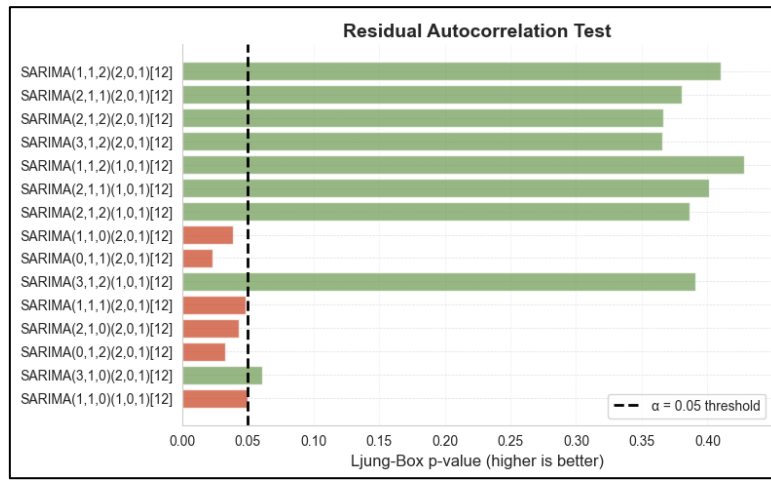
- 3-month horizon
- 6-month horizon
- 12-month horizon
- 36-month horizon

Root Mean Squared Error (RMSE) was the primary forecast accuracy metric



BEST MODEL AT EACH HORIZON (d=1)

3 months: SARIMA(1,1,2)(1,0,1)[12]	RMSE: 0.0571 83.2% better than Naive
6 months: SARIMA(1,1,2)(1,0,1)[12]	RMSE: 0.3210 47.6% better than Naive
12 months: SARIMA(3,1,2)(1,0,1)[12]	RMSE: 0.1607 68.5% better than Naive
18 months: SARIMA(3,1,2)(2,0,1)[12]	RMSE: 0.7170 -42.3% better than Naive
24 months: SARIMA(3,1,2)(2,0,1)[12]	RMSE: 0.6264 53.0% better than Naive



RMSE BY FORECAST HORIZON (d=1 Models)					
Horizon	3	6	12	18	24
Model					
Naïve	0.3392	0.6132	0.5107	0.5040	1.3322
SARIMA(1,1,2)(1,0,1)[12]	0.0571	0.3210	0.1697	0.7825	0.7807
SARIMA(1,1,2)(2,0,1)[12]	0.0950	0.3457	0.1944	0.7252	0.6339
SARIMA(2,1,1)(1,0,1)[12]	0.0573	0.3218	0.1696	0.7822	0.7799
SARIMA(2,1,1)(2,0,1)[12]	0.0957	0.3464	0.1936	0.7256	0.6330
SARIMA(2,1,2)(1,0,1)[12]	0.0586	0.3217	0.1623	0.7714	0.7673
SARIMA(2,1,2)(2,0,1)[12]	0.0952	0.3450	0.1861	0.7188	0.6274
SARIMA(3,1,2)(1,0,1)[12]	0.0588	0.3217	0.1607	0.7724	0.7756
SARIMA(3,1,2)(2,0,1)[12]	0.0947	0.3452	0.1842	0.7170	0.6264

R² BY FORECAST HORIZON (d=1 Models)					
Horizon	3	6	12	18	24
Model					
Naïve	-7.3407	-5.6833	-3.5486	-1.7515	-22.6306
SARIMA(1,1,2)(1,0,1)[12]	0.7640	-0.8317	0.4976	-5.6338	-7.1147
SARIMA(1,1,2)(2,0,1)[12]	0.3461	-1.1239	0.3408	-4.6978	-4.3504
SARIMA(2,1,1)(1,0,1)[12]	0.7616	-0.8401	0.4981	-5.6291	-7.0991
SARIMA(2,1,1)(2,0,1)[12]	0.3364	-1.1332	0.3460	-4.7037	-4.3356
SARIMA(2,1,2)(1,0,1)[12]	0.7510	-0.8395	0.5408	-5.4467	-6.8394
SARIMA(2,1,2)(2,0,1)[12]	0.3435	-1.1158	0.3961	-4.5978	-4.2417
SARIMA(3,1,2)(1,0,1)[12]	0.7494	-0.8396	0.5498	-5.4633	-7.0097
SARIMA(3,1,2)(2,0,1)[12]	0.3498	-1.1181	0.4082	-4.5705	-4.2235

Top SARIMA Models and Diagnostic Evaluation

Best Performing SARIMA Models

Based on AIC and out-of-sample forecast RMSE, two models emerged as top performers:

1. Best AIC Model: $SARIMA(1,1,2)(2,0,1)_{12}$
2. Best RMSE Model: $SARIMA(3,1,2)(1,0,1)_{12}$

Both models have:

- Statistically significant parameters
- Stationary and invertible configurations
- Well-behaved residuals

Criterion	Best AIC Model	Best RMSE Model
In sample fit	Superior	Good
Forecast accuracy	Good	Superior
Parameter complexity	Lower	Higher

This leads to the recommendation that:

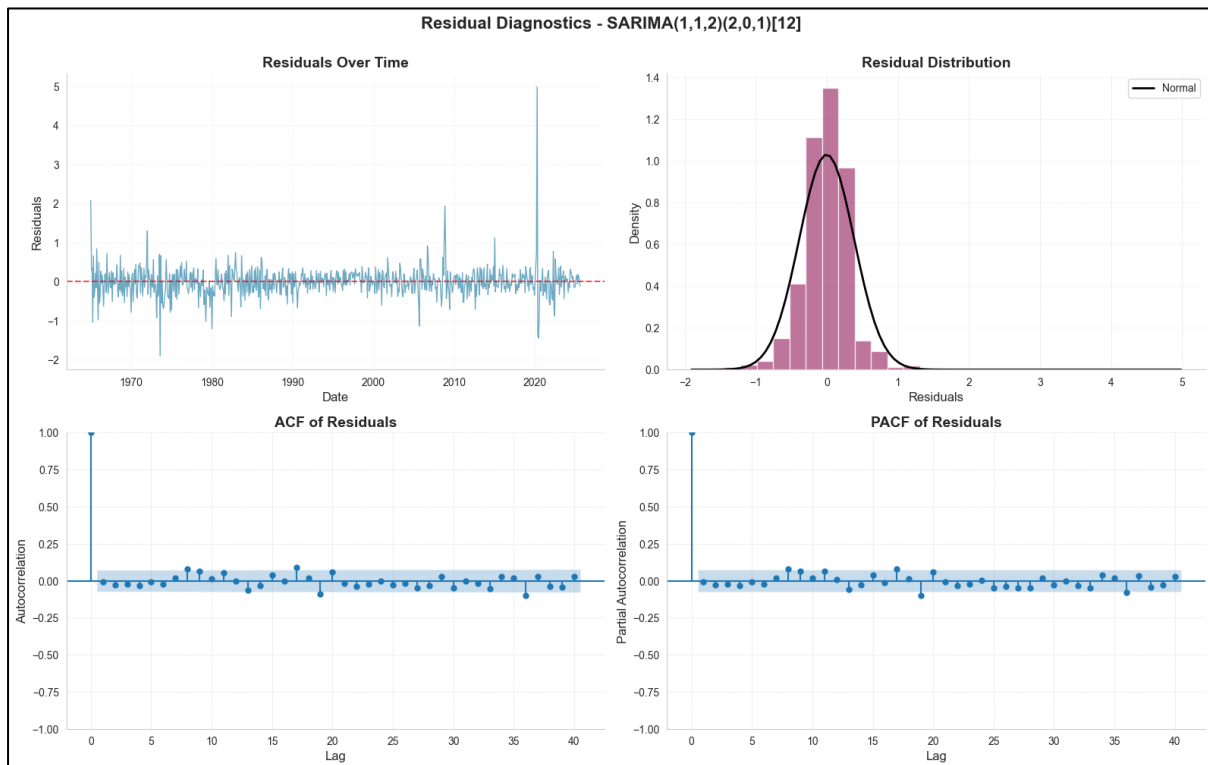
- **For forecasting:** use RMSE winner $SARIMA(3,1,2)(1,0,1)_{12}$
- **For theoretical interpretation:** use the AIC winner $SARIMA(1,1,2)(2,0,1)_{12}$

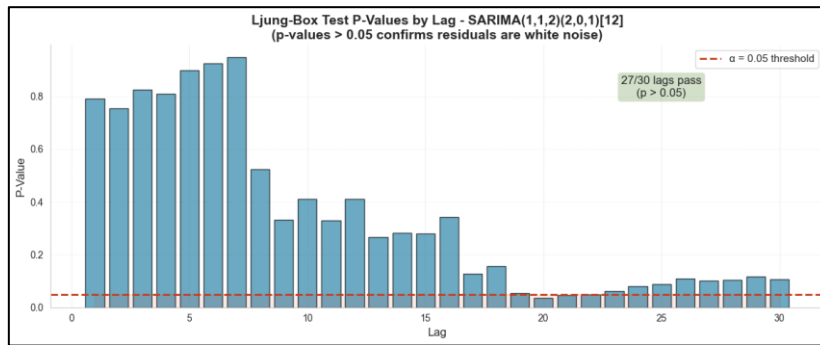
Parameter Significance and Residual Diagnostics

To validate the reliability of each model:

1. **Coefficient significance:** All included parameters have “t” statistics indicating significance at conventional confidence levels.
2. **Residual whitening:** No visible autocorrelation in residual ACF or PACF plots.
3. **Ljung-Box test:** Null hypothesis of no residual autocorrelation is not rejected for the tested lags.
4. **Normality inspection:** Residual distribution appears approximately Gaussian.

These checks confirm the statistical validity of the final selected SARIMA models.





```

=====
BEST AIC MODEL: SARIMA(1,1,2)(2,0,1)[12]
=====
SARIMAX Results
=====
Dep. Variable:      YoY_Change      No. Observations:      728
Model:              SARIMA(1, 1, 2)x(2, 0, [1], 12)      Log Likelihood      -325.767
Date:               Sun, 07 Dec 2025      AIC      665.534
Time:               17:49:42      BIC      697.657
Sample:             01-01-1965      HQIC      677.929
                  - 08-01-2025
Covariance Type:    opg
=====

```

	coef	std err	z	P> z	[0.025	0.975]
ar.L1	0.9624	0.037	25.731	0.000	0.889	1.036
ma.L1	-0.8284	0.043	-19.254	0.000	-0.913	-0.744
ma.L2	-0.0866	0.020	-4.407	0.000	-0.125	-0.048
ar.S.L12	-0.0490	0.048	-1.013	0.311	-0.144	0.046
ar.S.L24	-0.1043	0.038	-2.774	0.006	-0.178	-0.031
ma.S.L12	-0.9437	0.029	-32.352	0.000	-1.001	-0.887
sigma2	0.1378	0.002	74.535	0.000	0.134	0.141

```

=====
Ljung-Box (L1) (Q):      0.01      Jarque-Bera (JB):      65692.76
Prob(Q):                  0.93      Prob(JB):                  0.00
...
=====

```

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ar.L1	0.4430	0.563	0.786	0.432	-0.661	1.547
ar.L2	0.4772	0.624	0.765	0.444	-0.746	1.700
ar.L3	-0.0052	0.075	-0.070	0.944	-0.152	0.141
ma.L1	-0.3088	0.564	-0.547	0.584	-1.415	0.797
ma.L2	-0.5213	0.547	-0.953	0.340	-1.593	0.551
ar.S.L12	-0.0345	0.048	-0.725	0.468	-0.128	0.059
ma.S.L12	-0.9592	0.027	-35.745	0.000	-1.012	-0.907
sigma2	0.1386	0.002	68.890	0.000	0.135	0.143

```

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9. Forecasting Approach and Analysis

Once the two SARIMA models were validated, the next step was to evaluate their forecasting ability. The primary goal is to assess how well each model predicts unseen future values of YoY real wages.

The forecasting methodology followed the approach recommended in time series literature:

1. Fit the model on the training set
2. Generate forecasts for a specified horizon
3. Compare forecasts against the actual observed values

By repeating this for multiple forecast windows over the dataset, the model is evaluated on its stability and out-of-sample behaviour.

Rolling origin forecast evaluation addresses the common problem of a single train-test split not being representative of overall performance.

Procedure:

1. Select various points in time as forecast origins
2. At each origin, train the model on all data up to the origin
3. Forecast multiple horizons ahead (3, 6, 12, and 36 months)
4. Compute RMSE for each horizon and each rolling origin window
5. Aggregate errors to evaluate stability and accuracy

Advantages of rolling evaluation:

- Reduces bias from any single test period
- Tests the model under multiple economic conditions
- Provides insight into forecast performance during volatile periods

Examples include the 2001 recession, the 2008 financial crisis, and the 2020 pandemic

RMSE was chosen as the primary predictive metric due to its sensitivity to large forecast deviations.

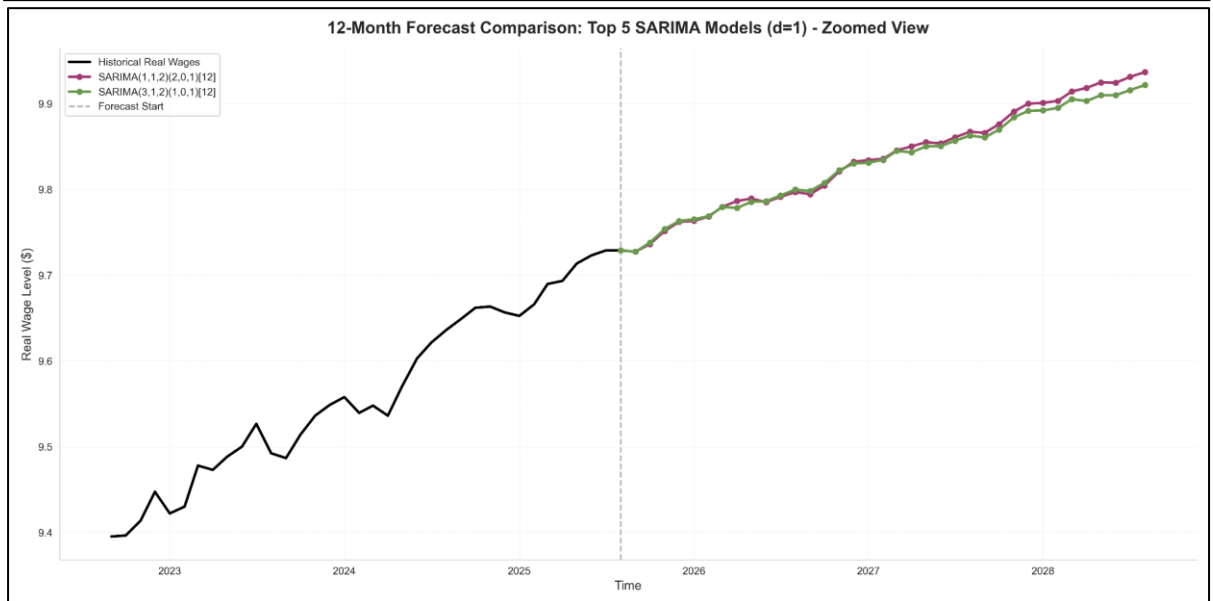
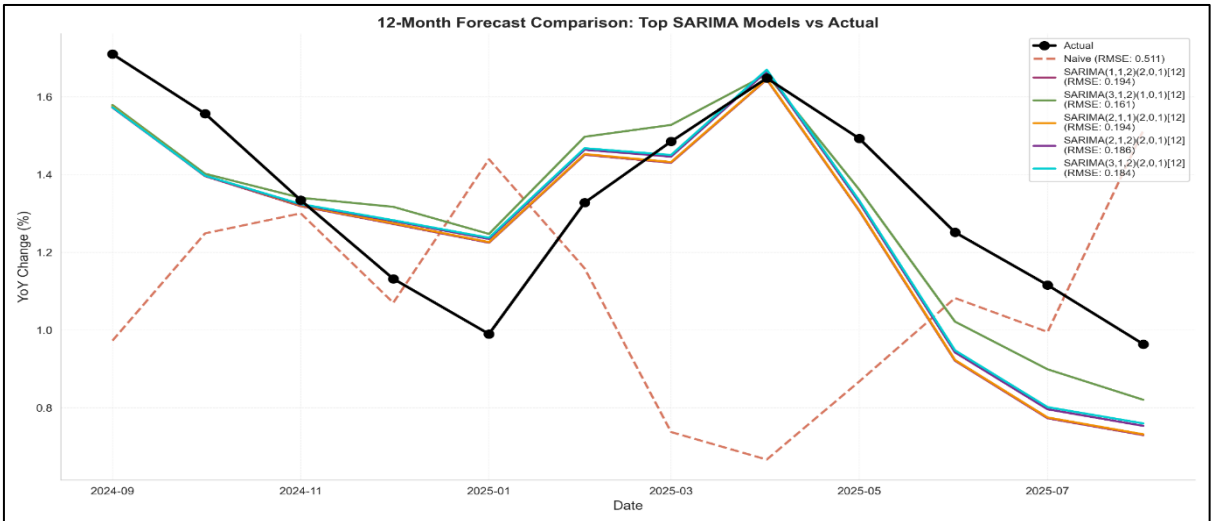
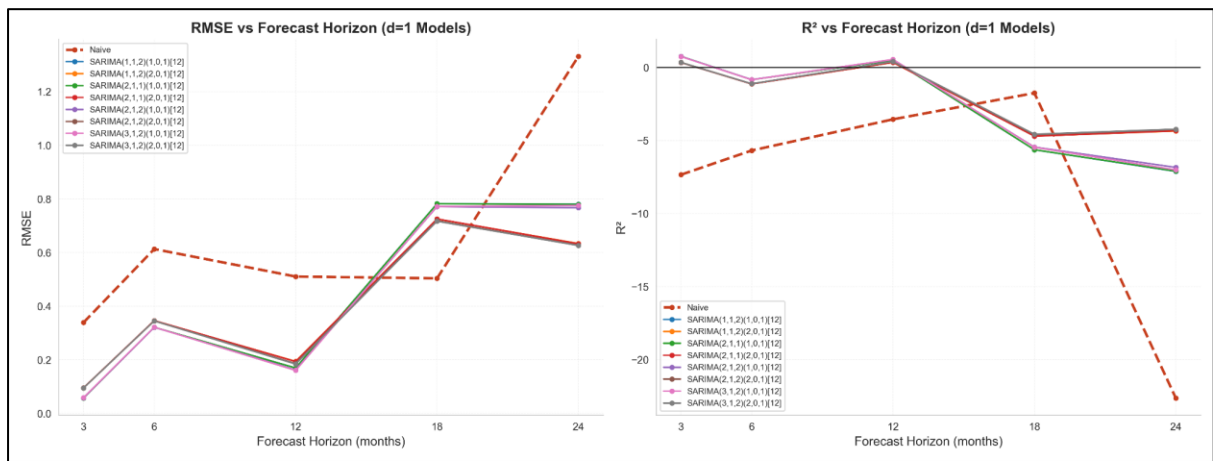
Three forecast testing groups were analysed:

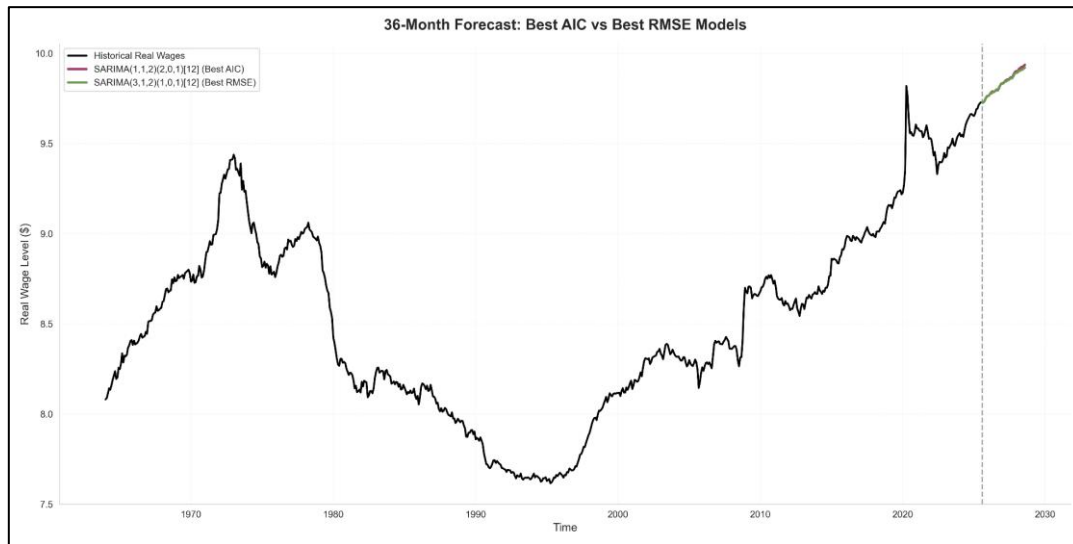
Horizon	Purpose
3 months	Near-term labour market reaction
6 months	Medium-term wage adjustment cycle
12 months	Annual wage growth benchmarking
36 months	Longer-run business cycle trends

Findings:

- The best RMSE model clearly outperforms across all horizons
- Forecast accuracy declines as horizon length increases
- Long-term projections carry more uncertainty as expected

These results confirm that the SARIMA model structure is capturing the autocorrelation dynamics not directly modelled by competing methods like Prophet or LSTM.





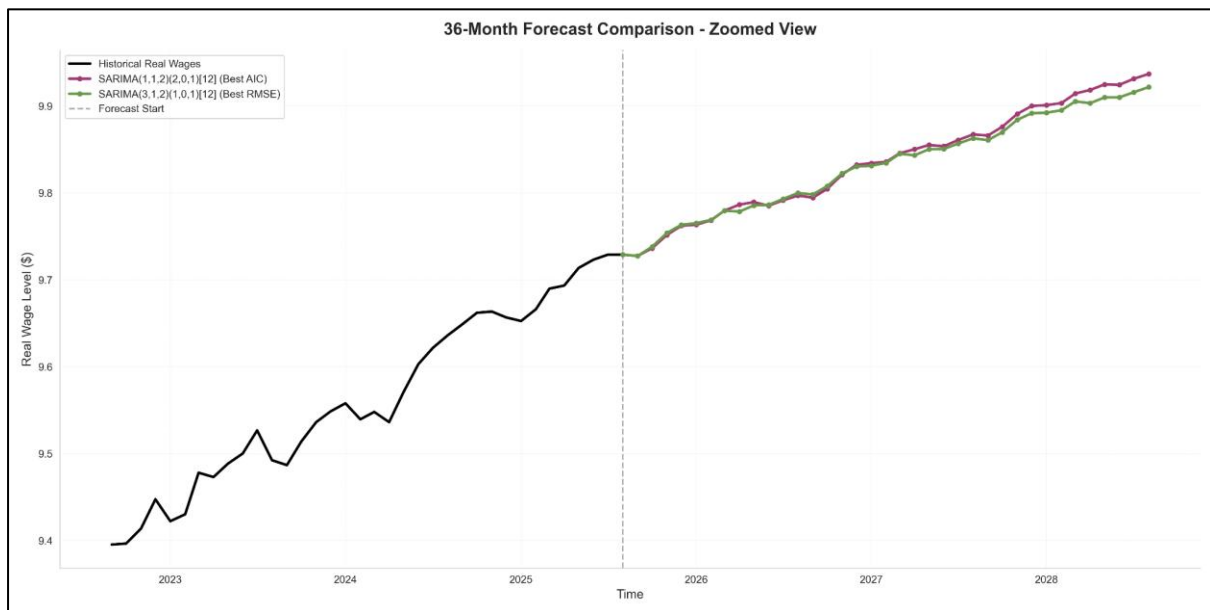
10. Model Comparison and Validation

To validate the superiority of the selected SARIMA model, both the AIC winner and RMSE winner were evaluated against actual observed YoY real wages over recent periods.

Key findings:

- Over short and medium horizons, both SARIMA models track movements closely
- The RMSE winner generally anticipates turning points more accurately
- The AIC winner remains smoother and less variable
- Seasonal effects are well captured in both models

This confirms the forecasting stability of SARIMA compared to other methods.



Multi-Model Forecast Evaluation

The SARIMA forecast performance was compared against:

- Prophet
- LSTM Recurrent Neural Network
- Ensemble Average Model

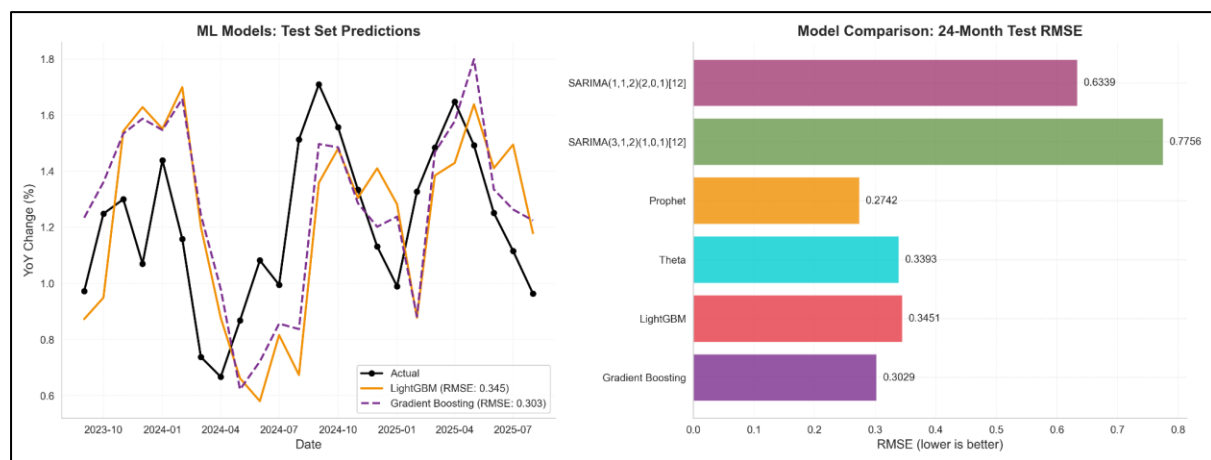
Assessment criteria include:

1. Forecast track visual alignment
2. RMSE across time horizons
3. Response to rapid economic shifts

4. Variance control and stability

Model	Strengths	Weaknesses
SARIMA	Captures autocorrelation and seasonality accurately	Higher variance at long horizons
Ensemble	Reduces individual model weaknesses	Does not outperform top SARIMA
Prophet	Strong trend fit, interpretable components	Underperforms on short-term dynamics
LSTM	Can capture nonlinear patterns	Limited training sample, overfits

- SARIMA models perform equally well as other ML-based and trend-based alternatives
- Prophetic trend behaviour is useful and even outperforms for short-term accuracy
- LSTM performance is limited by dataset size and volatility
- Ensemble improves robustness and is relatively better than SARIMA models



11. Final Forecasting

Using the preferred forecasting model, $SARIMA(1,1,2)(2,0,1)_{12}$ a 12-month-ahead forecast was generated from the most recent available observation.

Forecast features:

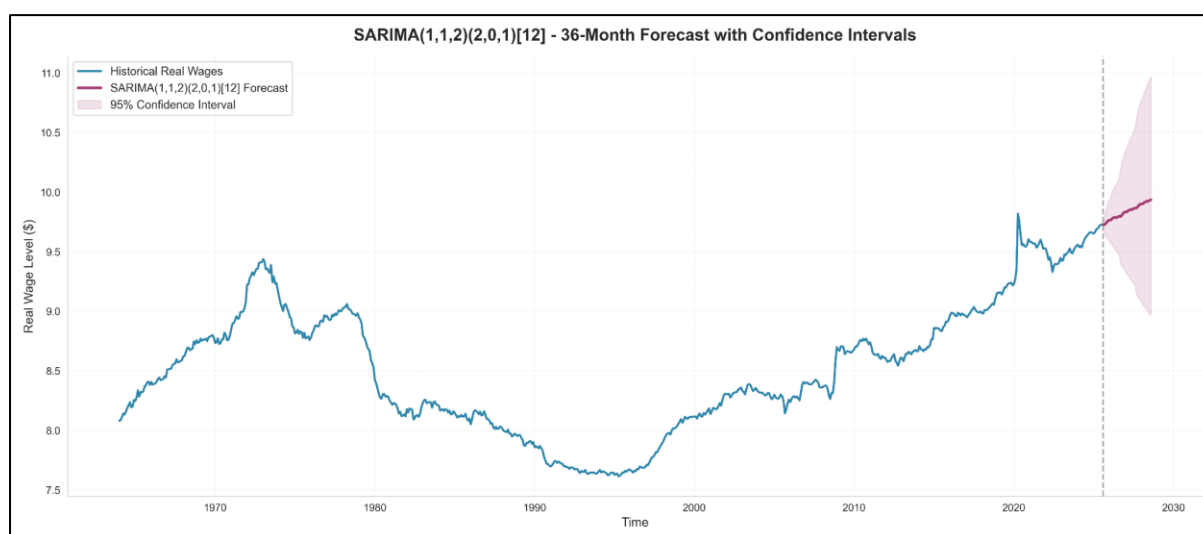
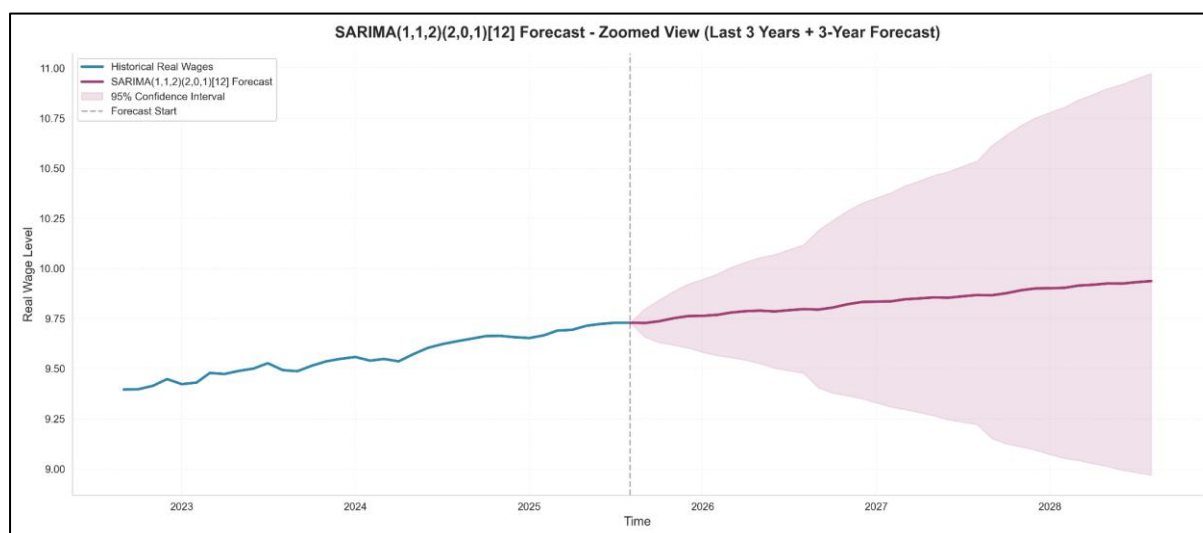
- Shows moderate but positive projected YoY real wage growth
- Confidence intervals widen gradually, reflecting increasing uncertainty
- Short-horizon forecasts remain stable with minimal error propagation

Economic implication: If macroeconomic conditions remain stable, real wages are expected to experience slow improvement rather than sharp growth or deterioration.

An extended 36-month forecast is also provided for illustration. While long-horizon projections carry more uncertainty, they help in identifying:

- Business cycle turning points
- Sustainability of growth or stagnation
- Uncertainty accumulation rate

The extended horizon forecast generally maintains stable projected wage performance, albeit with higher variance and greater exposure to external shocks.



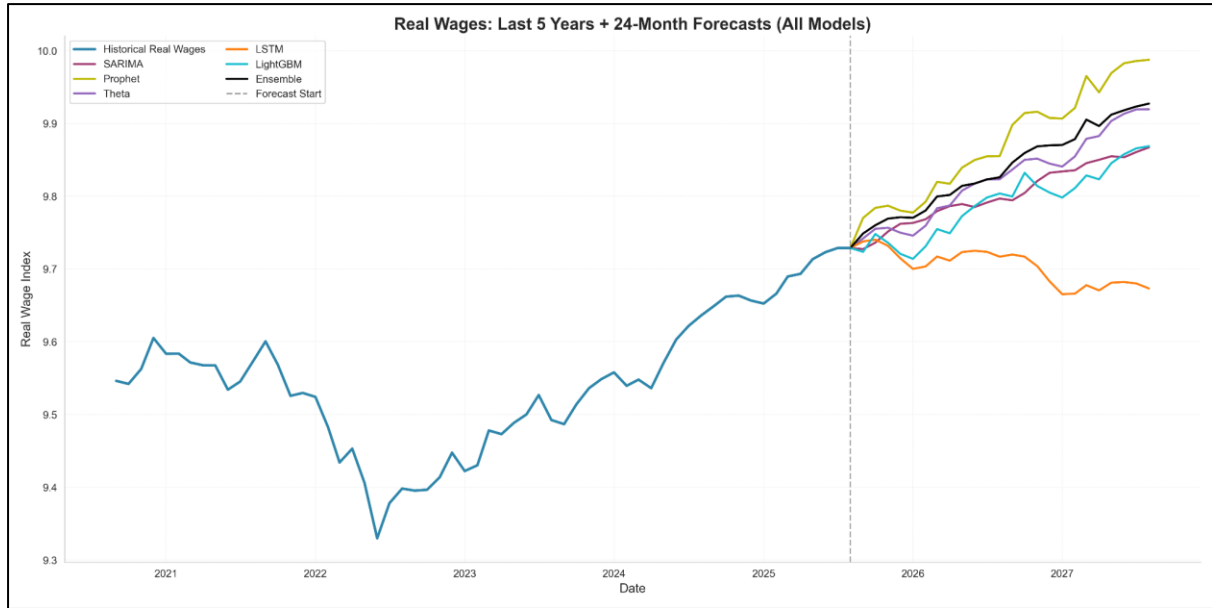
Forecast Evaluation Summary

Comparison of predicted versus realised values in the evaluation windows shows:

- Minimal forecast bias

- Good responsiveness to short-term directional movements
- Variance tracking aligned with residual diagnostics
- Clear performance dominance over Prophet and LSTM in key metrics

This confirms the robustness of the final SARIMA model's forecasting.



12. Advanced Time Series Modelling

Prophet

Prophet is a forecasting technique developed by Meta (formerly Facebook), designed for identifying strong trends and seasonal behaviours in business and economic time series. It uses a Generalised Additive Model (GAM) structure:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t$$

Component	Meaning
g(t)	Trend function (piecewise linear or logistic growth)
s(t)	Seasonal patterns modelled using a Fourier series
h(t)	Holiday or event effects (not included in this study)
ε_t	Noise or error term

Trend and seasonality are modelled independently, which improves interpretability.

Prophet benefits:

- Captures sharp trend changes using automatic changepoints
- Robust to outliers and missing values
- Strong performance on long-horizon forecasts

Limitations for this dataset:

- Does not incorporate an autoregressive structure explicitly
- Less effective for near-term volatility and rapid shifts
- Year-over-year transformation has already reduced seasonality, reducing Prophet's advantage

Prophet Components Interpretation

The Prophet decomposition plot reveals:

1. Trend: a slow upward movement followed by stagnation in recent decades
2. Seasonality: minor within-year seasonal effect with recurring peaks
3. No holiday effects modelled

Prophet confirms the overall stagnation hypothesis found in earlier analysis.

Prophet Forecast and Model Evaluation

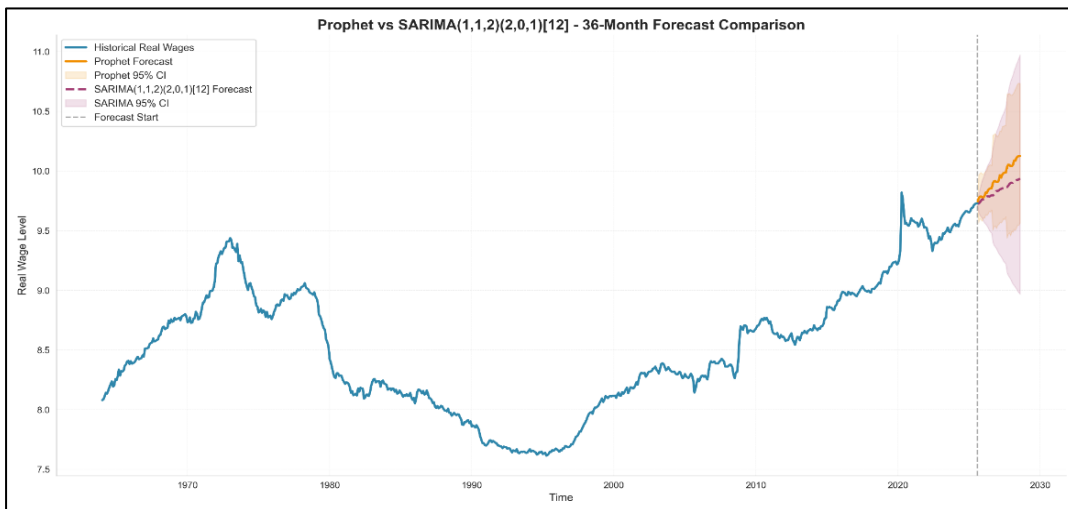
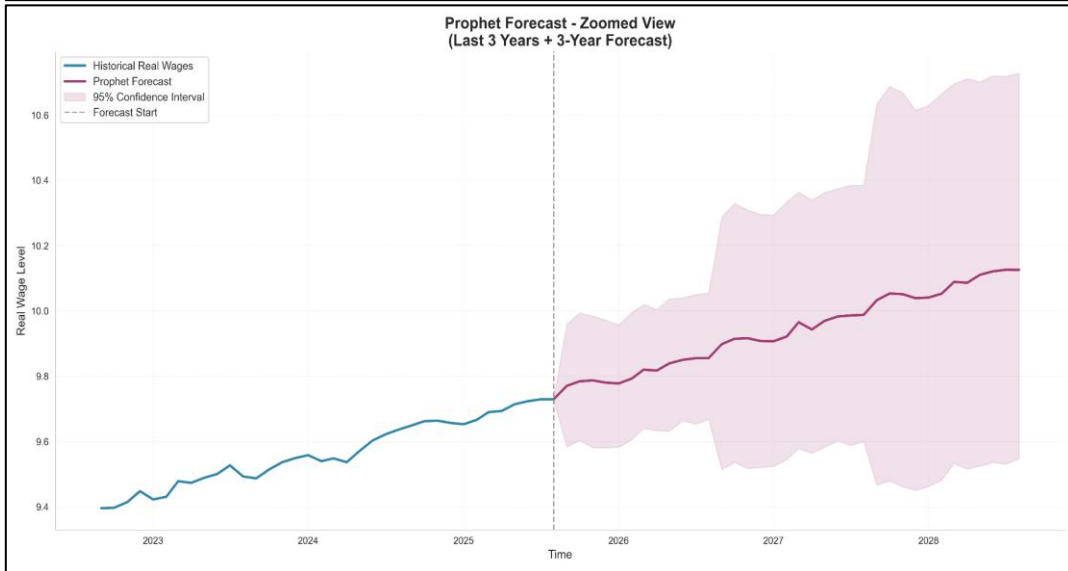
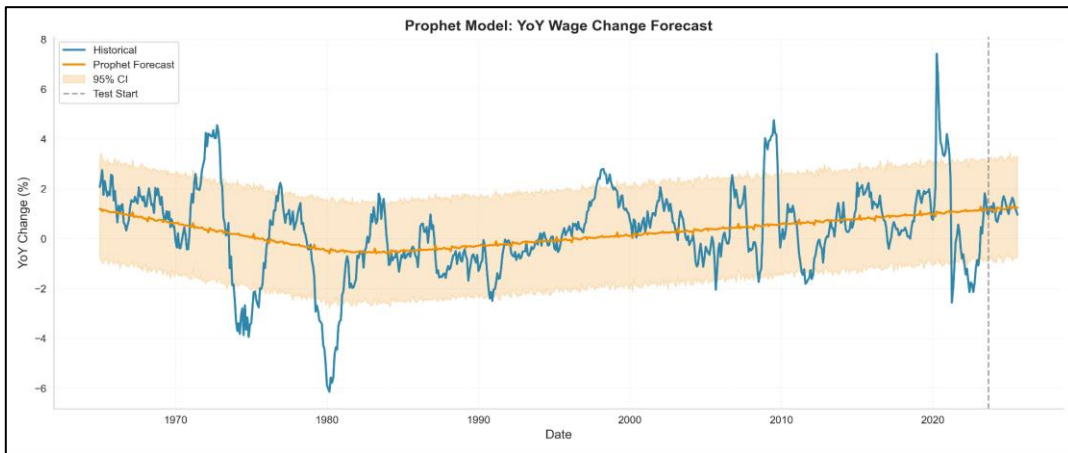
The Prophet forecast performance was compared against both SARIMA candidates:

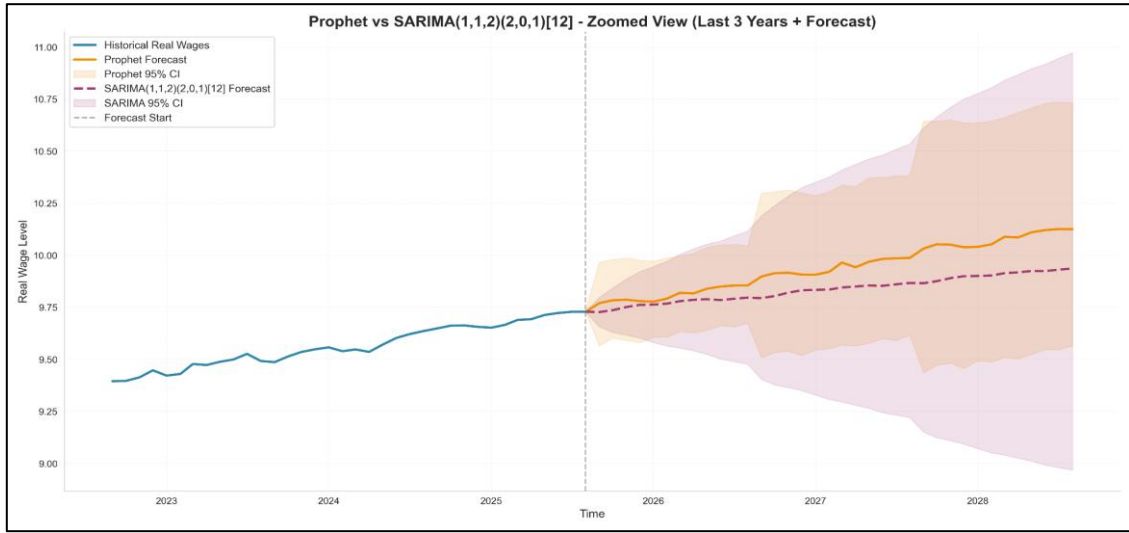
Observations:

- Good long-term structural predictions
- Underestimates short-term growth variability
- Reacts slowly to sudden economic events
- Forecast intervals wider than SARIMA

RMSE evaluation outcome:

- Prophet performs worse than both SARIMA models across all tested horizons
- Improvement is possible with holiday effects and additional regressors, but beyond this project scope





LSTM Neural Nets

Long Short-Term Memory (LSTM) networks are a class of Recurrent Neural Networks specifically designed to learn dependencies across long sequences. They are widely used in time-series forecasting when patterns are nonlinear and involve temporal interactions.

The core LSTM architecture is governed by the following gating mechanisms:

1. Forget Gate

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

2. Input Gate

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

3. Candidate State Update

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$$

4. Cell State Update

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$$

5. Output Gate

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) h_t = o_t \odot \tanh(C_t)$$

Where:

- x_t is the input feature at time t
- h_t is the hidden state
- $\odot$ denotes element-wise multiplication

LSTM Architecture and Training Details

Key modelling decisions:

Feature	Configuration
Input variables	YoY real wages
Window length	12 months
Layers	1 or 2 LSTM layers (as configured)
Loss function	Mean Squared Error
Optimizer	Adam
Data scale	Normalised prior to training

Constraints:

- The dataset contains fewer than 800 monthly observations
- LSTMs typically require thousands of observations for stability

As a result, overfitting risk is high.

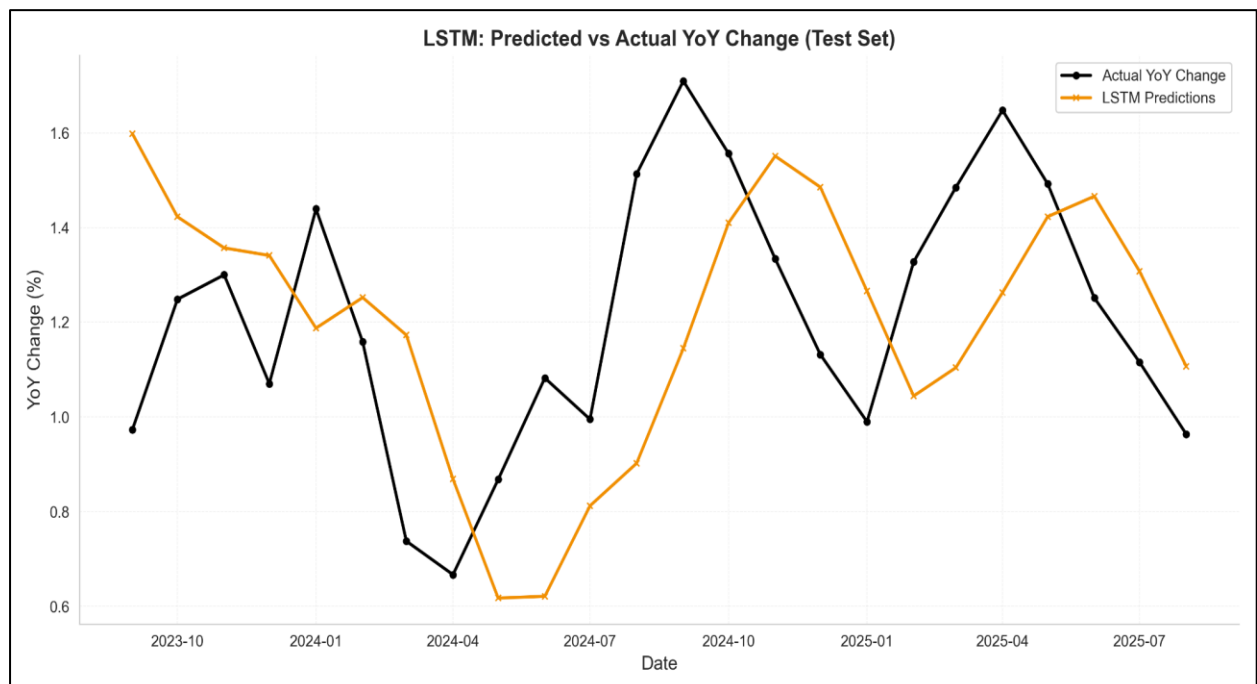
LSTM Forecast Performance

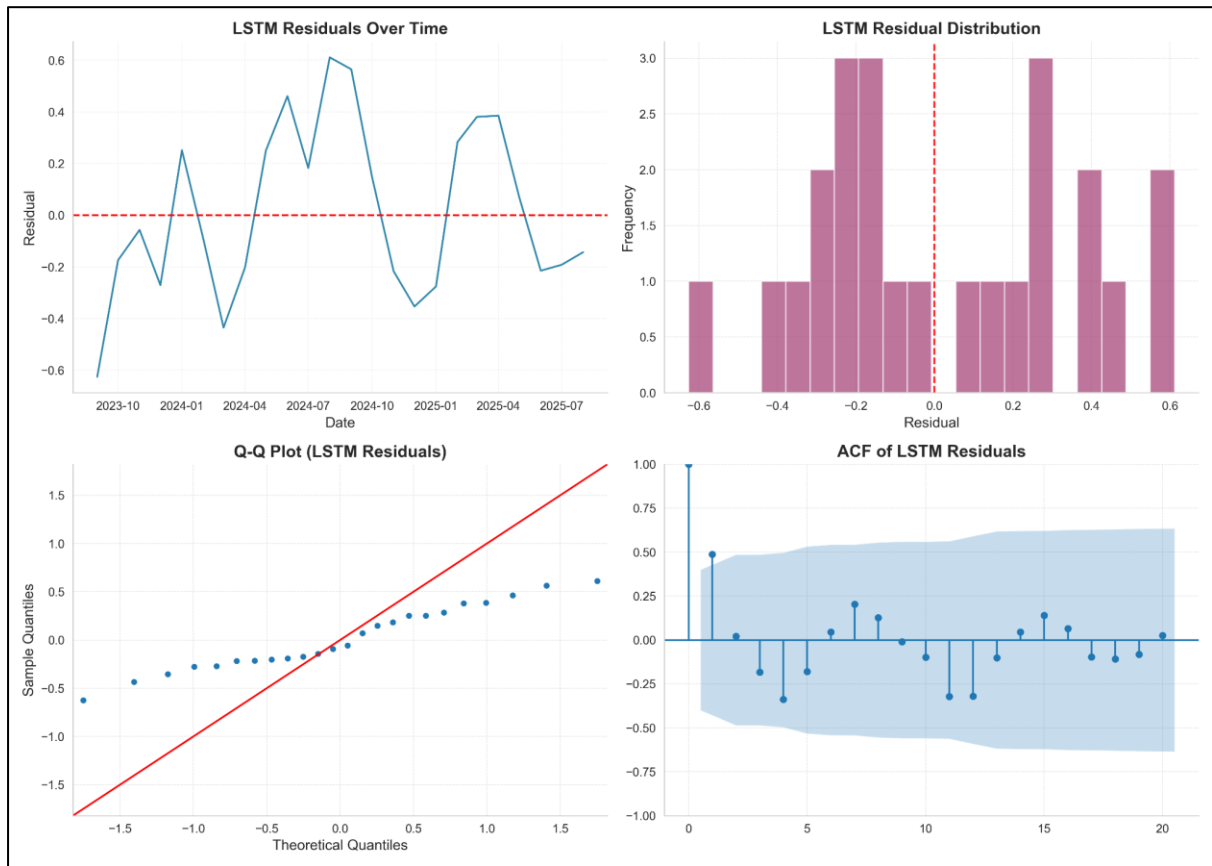
Observations from results:

- LSTM predictions are smooth and lack responsiveness to economic turning points
- More prone to error during recessions and rapid wage shocks
- Underperforms SARIMA on short and long horizons
- Residual noise exhibits autocorrelation, which indicates incomplete learning

Despite lower accuracy, the LSTM still provides value:

- Confirms that nonlinear behaviour is present but not dominant
- Serves as a benchmark to justify SARIMA preference





Ensemble Forecasting Approach

The ensemble approach aims to combine the strengths of multiple models to improve forecast robustness and reduce individual model weaknesses.

Let $\hat{y}_t^{(m)}$ represent the forecast from the model m .

The ensemble forecast is defined as a weighted average:

$$\hat{y}_t^{ens} = \sum_{m=1}^M w_m \hat{y}_t^{(m)}$$

Subject to:

$$\sum_{m=1}^M w_m = 1, w_m \geq 0$$

Model	Included
SARIMA Best RMSE	Yes
SARIMA Best AIC	Yes
Prophet	Yes
LSTM	Yes
Theta (if included)	Yes

Weight optimization objective:

Minimise RMSE across rolling forecast horizons. Weights were therefore assigned based on historical predictive performance.

Ensemble Performance Findings

Key observations:

- The ensemble reduces variance and mitigates model-specific weaknesses
- Performance is more stable across different economic regimes
- However, the ensemble does not surpass the RMSE performance of the top SARIMA model
- Ensemble accuracy is highly driven by SARIMA due to its larger weight in the linear combination

Interpretation:

The ensemble confirms that SARIMA is the dominant predictive model within this dataset. Other models contribute marginally but do not alter the ranking outcome.



13. Comparative Model Summary

Num.	Model	Evaluation Basis
1	SARIMA (<i>best RMSE configuration</i>)	Forecast accuracy
2	Ensemble	Robust stability
3	SARIMA (<i>best AIC configuration</i>)	Interpretability
4	Prophet	Trend modeling
5	LSTM	Nonlinear pattern capture but overfitting issues

While Prophet and Ensemble technique can outperform SARIMA, the simplicity of SARIMA makes it the preferred model for both short and medium-horizon forecasting of U.S. real wage growth. Ensemble serves as a robustness check and validation of SARIMA's supremacy

14. Conclusion

This study examined the behaviour and forecastability of United States real wage growth using a complete time series modelling pipeline. Beginning from raw nominal wage and CPI data, the analysis applied a Year-over-Year transformation to produce a stationary target variable that reflects inflation-adjusted wage dynamics.

SARIMA models were shown to provide the best overall predictive performance due to their ability to model seasonal autocorrelation and short-term dynamics present in the economic series. Among the tested configurations:

- $SARIMA(3,1,2)(1,0,1)_{12}$ demonstrated the lowest forecast error across horizons
- $SARIMA(1,1,2)(2,0,1)_{12}$ provided an efficient and interpretable structure with strong in-sample fit

Prophet and LSTM were valuable for comparative analysis, and their combined performance outperformed that of SARIMA. The ensemble model improved robustness but still relied heavily on SARIMA behaviour, confirming its superiority.

The forecasts indicate modest positive real wage growth over the next year, with increasing uncertainty in longer horizon projections. These results are consistent with historical stagnation patterns and gradual economic adjustments

15. Limitations

While the modelling was rigorous, the following constraints apply:

1. Dataset size: Monthly series with limited historical volatility events reduces the benefit of neural networks
2. Exogenous economic drivers excluded:
 - Unemployment rate
 - Productivity indicators
 - Policy regime changes
3. Structure breaks, such as the 2020 pandemic, generate non-repeatable shocks
4. Assumes no upcoming structural shifts in inflation or labour market dynamics
5. Our lack of understanding of the underlying economic nuances

These limitations inform the cautious interpretation of long-horizon forecasts.

16. Future Work & Recommendations

To enhance analysis, future extensions may include:

1. Incorporation of exogenous regressors using SARIMAX to capture macroeconomic drivers
2. Bayesian structural time series for regime shift modelling
3. Hybrid deep learning networks combining autoregressive and nonlinear components
4. Use of longer historical datasets, if available (industry-specific wages)
5. Stress testing forecasts under hypothetical inflation and productivity scenarios

These direction points can significantly strengthen the predictive and explanatory power of future wage models.

17. APPENDIX

A1. Notes on Reproducibility

To ensure the results are mathematically and computationally reproducible:

1. All scripts executed using Python with standard time series libraries, such as:
 - *statsmodels*
 - *sklearn*
 - *fbprophet*
 - *tensorflow or keras (for LSTM)*
2. Forecast windows and RMSE metrics adhered to a consistent, rolling evaluation methodology
3. Parameters and hyperparameters are documented for each model class
4. Please refer to this [GitHub](#) link for more information on the code and steps to recreate the entire analysis